

WASP-2b

R-1516

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_210630 · created 2026-08-01T16:32:12+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459395.754 +/- 0.0067 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.158 +/- 0.045
Transit depth (Rp/Rs)^2	2.49 +/- 1.43 [%]
Orbital Inclination (inc)	86.2 +/- 2.91
Airmass coefficient 1 (a1)	1.08 +/- 0.8
Airmass coefficient 2 (a2)	-0.09 +/- 0.17
Scatter in the residuals of the lightcurve fit is	67.4 %
Best Comparison Star	#1 - [538, 304]
Optimal Aperture	2.14
Optimal Annulus	4.77
Transit Duration (day)	0.077 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.158 ± 0.045 vs 0.1326 (deviation 0.56σ)
Duration fit vs published	0.077 d vs 0.0737 d (deviation 0.12σ)
Mid-time O–C	13.3 min
Depth SNR	3.5
Residual scatter	67.4 %

Light curve

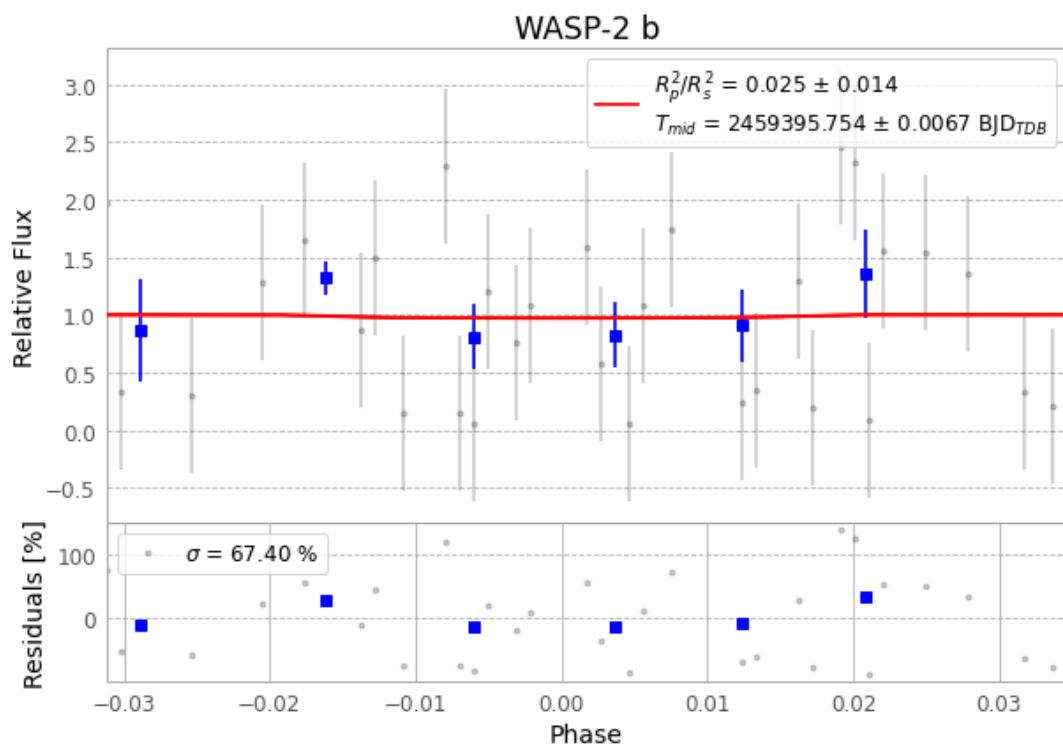


Figure 1 — FinalLightCurve_WASP-2 b_2021-06-29.png (SHA-256 0a1c6af028a677e1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [610, 69], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 010bd06f3215f88c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

69 FITS frames (45 MB), WASP-2210630042112.FITS → WASP-2210630074512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-210630.log (4144c12309e9...), FinalLightCurve_WASP-2 b_2021-06-29.png (0a1c6af028a6...), FinalLightCurve_WASP-2 b_2021-06-29.csv (8ee2a7915091...)

Compute resources

Wall time 195.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 16:32Z.